Class 6: The Ideal Gas

The theory that describes the behavior of an ideal gas is referred to as the Kinetic theory of gases. In this class we will briefly review the approach and some of the major predictions of the Kinetic theory of gases.

Let us consider a closed volume 'V', defined by a cubic box of length 'L', which contains 'n' moles of an ideal gas, as shown in Figure 6.1. Let a molecule of the ideal gas, of mass 'm', traveling in the positive x direction, collide with the wall of the box and bounce back. If we assume the collision is elastic, then change in momentum of the molecule is only in the x direction.

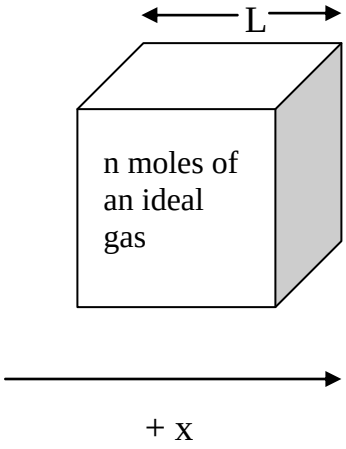


Figure 6.1: n moles of an ideal gas confined to a cubic box of length L

Initial momentum $+mv_x$

Final momentum $-mv_x$

Change in momentum = final momentum – initial momentum = $-mv_x - (+mv_x) = -2mv_x$

Given that the particle is traveling with $\pm v_x$, and the length of the box is *L*,

the time between collisions is $\frac{2L}{v_x}$ (since the particle has to traverse the distance '*L*' in each direction once before it can collide with the same wall again)

Therefore, the rate at which momentum is delivered to the wall (which is opposite to the rate of change of momentum for the particle) is given by:

$\frac{-(-2mv_x)}{2L/v_x} = \frac{mv_x^2}{L}$; which is then the force exerted by the particle on the wall

Considering all of the particles, the total force exerted on the wall is given by

$$F = \frac{mv_{x_1}^2}{L} + \frac{mv_{x_2}^2}{L} + \ldots + \frac{mv_{x_N}^2}{L}$$

The pressure exerted on the wall is given by:

$$P = \frac{F}{L^2}$$

Therefore

$$P = \frac{\sum_{i=1}^{i=N} mv_{x_i}^2}{L^3}$$

If we assume that the molecules have a mean square velocity in the x direction, denoted by $< v_x^2 >$, the pressure exerted on the wall can be written as:

$$P = \frac{\sum_{i=1}^{i=N} m < v_x^2 >}{L^3} = \frac{mN < v_x^2 >}{L^3}$$

Since there are 'n' moles of gas in the system, and 'N' is the total number of molecules present, $N = nN_A$, where N_A is the Avogadro number.

Therefore the pressure can be written as:

$$P = \frac{mnN_A < v_x^2 >}{L^3}$$

If the overall velocity of a particle is $\vec{v}$, then

$$v^2 = v_x^2 + v_y^2 + v_z^2$$

Since the gas molecules are in random motion, there is no preferred direction of motion, therefore we can reasonably assume that:

$v_x^2 = v_y^2 = v_z^2$

Therefore: $v_x^2 = \frac{v^2}{3} = \frac{v_{rms}^2}{3}$

Further, since the volume of the box, L^3, is equal to 'V'

The equation for pressure becomes:

$$P = \frac{mnN_A v_{rms}^2}{3V}$$

Rearranging, and comparing with the equation for an ideal gas, we obtain:

$PV = \frac{nmN_A v_{rms}^2}{3} = nRT$; where '*R*' is the universal gas constant and '*T*' is the absolute temperature

If "M' is the molar mass of the gas, then $M = mN_A$

Hence:

$$v_{rms}^2 = \frac{3RT}{M}$$

Or

$$\frac{Mv_{rms}^2}{3} = RT$$

Dividing by N_A on both sides, we obtain on a per molecule basis:

$\frac{mv_{rms}^2}{3} = k_b T$;

where k_b is the Boltzmann constant and is related to the universal gas constant *R* through $k_b = \frac{R}{N_A}$, where N_A is the Avogadro number.

Therefore:

$$\frac{1}{2}mv_{rms}^2 = \frac{3}{2}k_bT$$

The above expression gives the average translational kinetic energy per ideal gas atom/molecule in a system at equilibrium at temperature T and pressure P.

If there are N_f atoms per unit volume, the thermal energy per unit volume, which is stored as the kinetic energy of the atoms, is given by

$$E = \frac{3}{2}N_f k_b T$$

Therefore, the specific heat at constant volume, C_v^e, is given by:

$$C_V^e = \frac{\partial E}{\partial T} = \frac{3}{2}N_f k_b$$

The expressions obtained for the average translational kinetic energy of atoms of an ideal gas, and the specific heat at constant volume, will be made use of in subsequent classes.